

Disproof of Conjecture 000000000348

(twin partial quotients via divisor function)

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1 Conjecture

There exists a real number α whose continued-fraction partial quotients satisfy

$$a_n = a_{n+1} = \lfloor \tau(n) \rfloor \quad \text{for all } n \geq 1,$$

where $\tau(n)$ denotes the divisor function (the number of positive divisors of n); moreover α is given by an explicit series construction.

2 Disproof

Verdict: FALSE.

The constraints imposed by consecutive indices overlap in a single partial quotient: the condition at n fixes a_{n+1} to $\lfloor \tau(n) \rfloor$, while the condition at $n+1$ fixes the same quantity a_{n+1} to $\lfloor \tau(n+1) \rfloor$. Hence such an α can exist only if

$$\lfloor \tau(n) \rfloor = \lfloor \tau(n+1) \rfloor \quad \text{for every } n \geq 1.$$

Take $n = 4$ and $n = 5$:

$$\tau(4) = |\{1, 2, 4\}| = 3, \quad \tau(5) = |\{1, 5\}| = 2.$$

The condition at $n = 4$ forces $a_4 = a_5 = 3$; the condition at $n = 5$ forces $a_5 = a_6 = 2$. Thus $a_5 = 3$ and $a_5 = 2$ simultaneously, a contradiction.

Therefore no real number α satisfies the stated partial-quotient pattern, and in particular none is obtainable from an explicit series construction. The conjecture is false.

3 Remarks

- Since $\tau(n) \in \mathbb{N}$, the floor is immaterial: $\lfloor \tau(n) \rfloor = \tau(n)$.
- The conflict occurs even earlier: $n = 1, 2$ give $a_2 = \tau(1) = 1$ and $a_2 = \tau(2) = 2$. The argument above uses the registered attack pair $(n, n+1) = (4, 5)$.
- In general the condition forces $\tau(n) = \tau(n+1)$ for all n , i.e. a constant divisor function, which is absurd since $\tau(1) = 1 \neq 2 = \tau(2)$; any adjacent pair with differing τ yields a disproof.

4 Verification

The numeric claims $\tau(4) = 3$ and $\tau(5) = 2$ are recomputed by `reproduce.py` and formalized in Lean 4 (no Mathlib, zero axioms) in `lean4/Main.lean`, which proves

$$\neg \exists a : \mathbb{N} \rightarrow \mathbb{N}, \forall n, a_n = \tau(n) \wedge a_{n+1} = \tau(n).$$